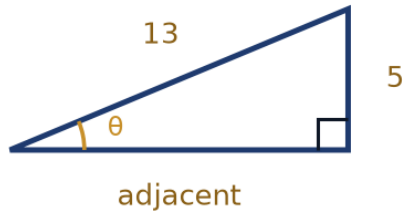


Trigonometry



Right-triangle trigonometric ratios

- 1 In a right triangle, angle θ has an opposite side of 5 and hypotenuse 13. Find $\sin \theta$ and the length of the adjacent side.



- 2 In a right triangle, angle θ has an opposite side of 15 and hypotenuse 17. Find $\cos \theta$.
- 3 In a right triangle, angle θ has an adjacent side of 7 and hypotenuse 25. Find $\sin \theta$ and $\tan \theta$.
- 4 In a right triangle, the two legs are 20 and 21, and the hypotenuse is 29. Find $\sin \theta$ and $\cos \theta$ for the angle θ opposite the side of length 20.

Special angles

- 5 State the exact value of $\sin(30^\circ)$.
- 6 State the exact value of $\cos(45^\circ)$.
- 7 State the exact value of $\tan(60^\circ)$.
- 8 State the exact values of $\sin(90^\circ)$ and $\cos(0^\circ)$.

Converting between radians and degrees

- 9 Convert 180° to radians.
- 10 Convert 90° to radians.
- 11 Convert $\frac{5\pi}{6}$ radians to degrees.
- 12 Convert 270° to radians.

The Pythagorean identity

- 13 Given $\sin \theta = \frac{3}{5}$ with θ acute, find $\cos \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$.
- 14 Given $\cos \theta = \frac{5}{13}$ with θ acute, find $\sin \theta$.
- 15 Given $\sin \theta = 0.6$, find $\cos^2 \theta$.

Simplifying trigonometric expressions

- 16 Simplify $\sin^2 \theta + \cos^2 \theta + 1$.
- 17 Simplify $\frac{1 - \cos^2 \theta}{\sin \theta}$ (for $\sin \theta \neq 0$).
- 18 Simplify $\tan \theta \cos \theta$ (for $\cos \theta \neq 0$).

Graphs: amplitude and period

- 19 State the amplitude and period of $y = 5\sin(x)$.
- 20 State the amplitude and period of $y = 3\cos(2x)$.
- 21 State the amplitude and period of $y = 2\sin\left(\frac{x}{3}\right)$.
- 22 A sine graph has amplitude 4 and period π . Write its equation in the form $y = A\sin(Bx)$.

Graphs: phase shift and vertical shift

- 23 The graph of $y = \sin(x)$ is shifted $\frac{\pi}{3}$ to the right and 2 units up. Write the new equation.
- 24 State the vertical shift and midline of $y = 3\cos(x) + 5$.
- 25 The graph of $y = \cos(x)$ is shifted $\frac{\pi}{4}$ to the left. Write the new equation.

Solving basic trigonometric equations

- 26 Solve $2\sin \theta = 1$ for $0^\circ \leq \theta < 360^\circ$.

27 Solve $\sqrt{2}\cos\theta = 1$ for $0^\circ \leq \theta < 360^\circ$.

28 Solve $\tan\theta = -1$ for $0^\circ \leq \theta < 360^\circ$.

29 Solve $2\cos\theta + 1 = 0$ for $0^\circ \leq \theta < 360^\circ$.

Reference angles in other quadrants

30 Find $\sin(150^\circ)$ using the reference angle.

31 Find $\cos(240^\circ)$ using the reference angle.

32 Find $\tan(300^\circ)$ using the reference angle.

Complementary angle relationships

33 Given $\sin(20^\circ) \approx 0.342$, use the relationship $\sin\theta = \cos(90^\circ - \theta)$ to find $\cos(70^\circ)$.

34 Given $\sin(35^\circ) \approx 0.574$, find $\cos(55^\circ)$.

The unit circle: coordinates of key points

35 State the coordinates $(\cos\theta, \sin\theta)$ of the point on the unit circle at $\theta = 60^\circ$.

36 State the coordinates of the point on the unit circle at $\theta = 180^\circ$.

37 State the coordinates of the point on the unit circle at $\theta = 270^\circ$.

Word problems: angles of elevation

38 This question has two parts. A drone hovers directly above a landmark. From a point 200 meters from the base of the landmark, the angle of elevation to the drone is 35° .

a Write an equation relating the drone's height h to the given information, using the tangent ratio.

b Find h , to the nearest meter, given $\tan(35^\circ) \approx 0.700$.

- 39** This question has two parts. A ladder 10 meters long leans against a wall, making an angle of 65° with the ground.
- a** Write an equation for the height h the ladder reaches up the wall, using the sine ratio.
 - b** Find h , to the nearest tenth of a meter, given $\sin(65^\circ) \approx 0.906$.